MEDINOVA TEACHING HOSPITAL

Class 08 — Neurology

Stroke · Seizures · Meningitis · Coma Assessment

> *"The brain does not announce its emergencies. You must find them."*

SENIOR RESIDENCY SERIES | MODULE 08 of 20

SECTION 01

Cranial Nerves & Motor Examination

Nerve	Name	Function (Test)	Abnormal Finding
I	Olfactory	Smell	Anosmia (Head trauma)
II	Optic	Acuity, Pupils, Fields	Hemianopia / Papilledema
III, IV, VI	Eye Move.	H-shape pursuit	Ptosis / Diplopia / Squint
V	Trigeminal	Facial sensation / Jaw	Absent corneal reflex
VII	Facial	Facial expressions	Bells Palsy (Forehead involved)
VIII	Vestibulo.	Hearing / Balance	Nystagmus / Vertigo
IX, X	Gloss./Vagus	Uvula shift / Gag reflex	Uvula deviates to *normal* side
XI	Accessory	Shrug shoulders	Weak Trapezius
XII	Hypoglossal	Stick out tongue	Tongue deviates to *affected* side

Motor/Sensory Framework

Tone: Hypertonia (Upper Motor Neuron) vs Hypotonia (Lower Motor Neuron).
Power (0-5 Scale): 0=No move; 3=Against gravity; 5=Normal.
Reflexes: 0=Absent; 2=Normal; 4=Clonus (UMN sign).

> **Prof. Obi's Pearl #1** "A patient who says 'I feel dizzy' is useless to you. You must differentiate Vertigo (spinning) from Syncope (passing out) or Disequilibrium (unsteadiness). They are three different worlds of pathology."

SECTION 02

Acute Ischaemic Stroke (AIS)

BE-FAST: Balance, Eyes (Visual loss), Face, Arms, Speech, Time.
Time is Brain: 1.9 million neurons are lost every minute during an AIS.

NIHSS Scoring Categories (0-42)

1. Consciousness | 2. Gaze | 3. Visual Fields | 4. Facial Palsy | 5. Motor (Arms/Legs) | 6. Ataxia | 7. Sensory | 8. Best Language | 9. Dysarthria | 10. Neglect.

Management Thresholds

Intervention	Time Window	Criteria / Contraindications
Thrombolysis (Alteplase)	< 4.5 Hours	BP must be < 185/110. No recent surgery/haemorrhage.
Thrombectomy (EVT)	< 6 - 24 Hours	Large Vessel Occlusion (LVO) on CT Angio.
Aspirin (300mg)	After 24h	Started only *after* CT rules out haemorrhage.

> **Prof. Obi's Pearl #2** "If the NIHSS score is 0 but the patient has a dense visual field defect, they are still a stroke candidate. Don't let a scoring system override your clinical eyes."

SECTION 03

Status Epilepticus Algorithm

Time (Mins)	Phase	Intervention & Dose
0 - 5	Stabilization	ABCDE, O2, Glucose check. Position safely.
5 - 20	Initial Therapy	Lorazepam 4mg IV (Repeat once) or Midazolam 10mg IM.
20 - 40	Second Phase	Levetiracetam (Keppra) 60mg/kg or Phenytoin 20mg/kg IV.
> 40	Third Phase	General Anaesthesia (Propofol/Thiopental) + Intubation.

SECTION 04

Bacterial Meningitis & CSF Analysis

Feature	Bacterial	Viral	TB / Fungal
Appearance	Turbid/Cloudy	Clear	Fibrin web
Glucose	Low (< 40% blood)	Normal	Low
Protein	High (> 1.0 g/L)	Normal / Mild ↑	Very High (> 1.0)
Cells	↑↑ Neutrophils	↑ Lymphocytes	↑ Lymphocytes

Rx: Ceftriaxone (2g IV BD) + Dexamethasone (10mg IV QDS - *must give before/with first Abx dose*).

SECTION 05

Coma Assessment & GCS

Eye Opening (4)	Verbal (5)	Motor (6)
4 - Spontaneous	5 - Oriented	6 - Obeys commands
3 - To Voice	4 - Confused	5 - Localizes pain
2 - To Pain	3 - Inappropriate	4 - Normal flexion
1 - None	2 - Incomp. sounds	3 - Abnorm. flexion
-	1 - None	2 - Extension / 1 - None

SECTION 06

Headache Red Flags (SNOOP4)

- **S:** Systemic symptoms (Fever/Weight loss).
- **N:** Neurological signs (Papilledema/Confusion).
- **O:** Onset sudden (Thunderclap = SAH).
- **O:** Older age (> 50 years).
- **P4:** Progression, Postural, Precipitated by Valsalva, Papilledema.

> **Prof. Obi's Pearl #3** "If a patient describes the 'worst headache of my life', do not wait for the CT results to start thinking about the Lumbar Puncture. But wait at least 12 hours post-onset for Xanthochromia to show up."

SECTION 07

Neurological Emergencies

1. Spinal Cord Compression:
• Back pain + Weakness + Bowel/Bladder dysfunction. Rx: High-dose Dexamethasone (16mg) + Urgent MRI within 24h.

2. Guillain-Barré Syndrome (GBS):
• Ascending paralysis + Areflexia. Danger: Respiratory failure. Rx: IVIG or Plasmapheresis. Avoid steroids!

3. Myasthenic Crisis:
• Severe weakness (Bulbar/Respiratory). Rx: Intubation + Plasmapheresis.

> **Prof. Obi's Pearl #4** "In neurology, the history gives you the diagnosis, the physical exam gives you the location, and the tests only confirm what you should already know."

Medical Definitions:
• **Xanthochromia:** Yellow tint of CSF indicating old blood (diagnostic for SAH).
• **Ataxia:** Lack of voluntary coordination of muscle movements.
• **Dysarthria:** Difficulty speaking due to muscle weakness.
• **Proprioception:** The body's ability to sense its location/movement.

OFFICIAL SIGN-OFF

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